BioScan-CLIP: Contrastive Learning for Aligning Biological Images and DNA Sequences

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Abstract

Measuring biodiversity is crucial for understanding ecosystem health. Whilst prior works have developed machine learning models for taxonomic classification of photographic images and DNA separately, in this work, we introduce a multimodal approach combining both, employing CLIP's methodology to align images, DNA barcodes, and textual data in a unified embedding space. This allows for accurate classification of both known and unknown insect species without task-specific fine-tuning, leveraging contrastive learning for the first time to fuse DNA and image data. Our method surpasses previous single-modality approaches in accuracy by over 11% on zero-shot learning tasks, showcasing its effectiveness in biodiversity studies.

Keywords: contrastive learning, biodiversity, DNA barcodes, machine learning

1. Introduction

With the accelerating pace of environmental change, monitoring biodiversity is crucial for maintaining ecosystem health. In recent years, advanced computer vision techniques have been used for species classification [1–4] and ecological monitoring [5], yet they may overlook species' evolutionary relationships and fine-grained differences. DNA sequencing, including DNA barcodes [6] like the COI gene, provides detailed species identification but is costly and equipment-intensive [7–11]. Thus, there is value in approaches that leverage DNA information at training time to build a model that solely requires images of new specimens at inference time [12].

Recent advances have enabled combining different data modalities. CLIP [13] and BioCLIP [14] have made strides in aligning images with textual information for classification. However, prior works only leverage a subset of the potential modalities rather than combining images, text, and DNA barcodes. Furthermore, they often expect a text description or complete taxonomic annotations for each sample, which are expensive and time-consuming to obtain.

In this work, we propose BIOSCAN-CLIP. By using contrastive learning to map biological images, textual taxonomic labels, and DNA barcodes to the same latent space, we can relax the constraint for comprehensive and correct taxonomic annotations. We can flexibly classify either images or DNA barcodes to predict the taxonomy, and the shared embedding space further enables work future research into commonalities and differences between species. In addition to the ecological benefits, building such a foundation model for biodiversity is a case study of a broader challenge to build models which can identify fine-grained differences, both visually and textually. Taxonomic classification is particularly interesting because those visual differences between species are often not well-defined, and the DNA and text modalities do not have much semantic overlap with natural language. We demonstrate the benefits of pretraining with all three modalities through improved taxonomic classification accuracy over prior works in both retrieval and zero-shot settings.

2. Method

To learn aligned representations of the three modalities (image, DNA barcode, and text), we apply LoRA [15] to fine-tune pretrained encoders for each with a multimodal contrastive loss (see Figure 1). During inference, we use our fine-tuned encoders to extract features for

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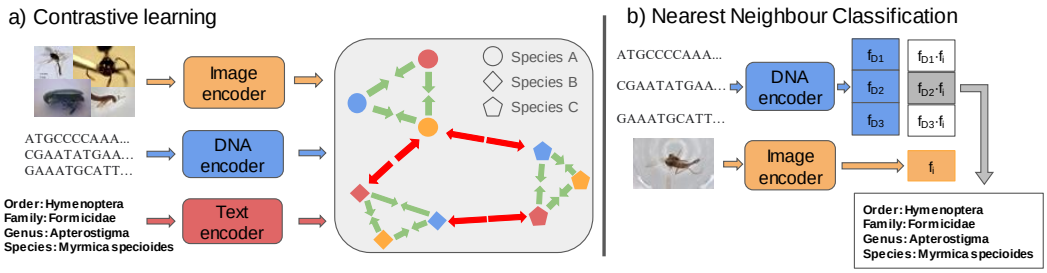


Figure 1. Overview of BIOSCAN-CLIP. **(a)** Our model consists of three encoders, dedicated to processing images, DNA barcodes, and text respectively. During training, we use a contrastive loss to align the image, DNA, and text embeddings. **(b)** At inference time, we embed a *query* image and match it to a database of existing image and DNA embeddings (*keys*). We use cosine similarity to find the closest key embedding and use its taxonomic label to classify the query image.

a *query* image and match against a database of image and DNA embeddings (*keys*). To classify taxonomy, we predict the taxonomic information associated with the most closely matched key. Querying against images and DNA comprise a more robust and defining set of records against which to query, given that the taxonomy is not always known or complete.

Contrastive Learning Following CLIP, we use the NT-Xent loss [16] between two modalities to align their representations. We extend CLIP based on [17] to three modalities by considering them in pairs. Let matrices $\mathbf{X}$, $\mathbf{D}$, and $\mathbf{T}$ represent the batch of ℓ_2-normalized embeddings of the image, DNA, and text modalities, respectively. The i-th row of each corresponds to the same instance, thus rows X_i and D_i comprise a positive pair of features from the same sample. Features in different rows X_i and D_j, $i \neq j$, are negative pairs. The contrastive loss for pair i is

$$L_i^{(X \to D)} = -\log \frac{\exp\left(X_i^T D_i/\tau\right)}{\sum_{k=1}^{n} \exp\left(X_i^T D_k/\tau\right)}, \quad L_{XD} = \sum_{i=1}^{n} \left(L_i^{(X \to D)} + L_i^{(D \to X)}\right)$$

where $\tau = 0.07$ is a fixed temperature, based on the initial value used in [13]. We apply the loss symmetrically to normalize over the possible paired embeddings for each modality [**zhang2021contrastive**, 17]. We repeat this for each pair of modalities and sum them to get the final loss, $L = L_{XD} + L_{DT} + L_{XT}$.

Pretrained encoders. For each data-modality, we use an encoder initialized as a pretrained model taken from the existing literature. *Images:* we use a ViT-B network pretrained on ImageNet-21k and fine-tuned on ImageNet-1k [18]. *DNA barcodes:* we use BarcodeBERT [11] with 5-mer tokenization, pretrained on a large corpus of DNA barcodes using masked language modelling. The training data for BarcodeBERT is highly similar to the DNA barcodes in BIOSCAN-1M, making it ideal for our study. *Text:* we use the pretrained BERT-small introduced by Turc, Chang, Lee, and Toutanova [19] to encode taxonomic labels.

Low-Rank Adaptation. We fine-tune pretrained transformer models using LoRA [15] by adding a low-rank residual layer to the query and key projectors of each attention module. The projectors take the form $W_{\text{LoRA}} = W + W_i W_o$, where $W \in R^{i \times o}$ represents the frozen pretrained weights, whilst $W_i \in R^{i \times r}$ and $W_o \in R^{r \times o}$ represent the added low-rank weights. By choosing a rank r much smaller than the input and output dimensions i and o, fewer parameters need to be updated with LoRA ($i\,r + r\,o < i\,o$).

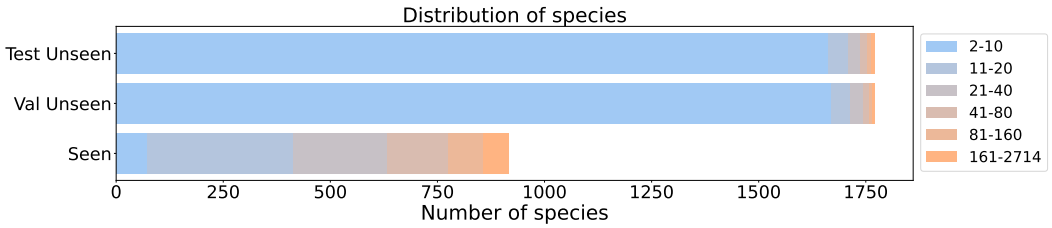


Figure 2. The distribution of species among the seen, validation unseen, and test unseen splits. Since all of the train, validation, and test seen splits share the same species, we represent them collectively. The coloured blocks represent the number of species with the respective numbers of records, demonstrating that most of the species in the unseen splits have few records.

3. Experiments

To evaluate our method, we perform taxonomic classification by using our model to match the input image (query) against the closest labeled DNA barcodes or other images (key). We assess accuracy across taxonomic levels by averaging over samples and taxon classes. Unlike basic fine-tuning with a fully connected layer, our approach identifies unseen species using images or DNA without knowing all potential species upfront. Instead, we use reference features for taxonomy prediction or novel class identification. We evaluate by designating some species as "unseen" during training and measuring prediction accuracy for both seen and unseen species to test model generalizability.

Dataset. For our training and experiments, we use the BioScan-1M dataset [20], a curated collection of over one million insect data records. Each record in the dataset includes a high-quality insect image, expert-annotated taxonomic labels, and a DNA barcode, making it well-suited for training our contrastive model. However, the dataset does not have complete taxonomic labelling for all specimens, with less than 10% of records labelled at the species level. This poses challenges for supervised training at these levels, but our methodology can nevertheless leverage this data effectively. The dataset also reflects the biological reality of a long-tail class imbalance, presenting additional challenges for our classification tasks, particularly in understanding how the different proportions of classes in the data set affect the accuracy of classification.

Data split. For our analysis, we split BIOSCAN-1M into training, validation, and test sets to evaluate zero-shot classification capabilities and model generalization to unseen species. Species with adequate records are divided 80/20 into *seen* and *unseen* categories for training/evaluation. Records lacking species labels are employed in contrastive pretraining, while species with 2–8 records are distributed to *unseen* splits in validation and test sets, ensuring *unseen* species are exclusive to their respective sets without overlap. This partitioning allows for a balanced distribution of *queries* and *keys* within *seen* and *unseen* splits for comprehensive model testing. See **??** for statistics on the splits.

Results. We conduct experiments to investigate how taxonomic classification accuracy changes depending on the modalities we aligned using contrastive learning to determine whether introducing DNA barcodes and taxonomic labels during training helps in learning improved features. We also assess the impact on accuracy with different data features as keys for prediction. By comparing the accuracy changes across different modalities, we can better understand the degree to which data from various modalities are aligned within the latent space. We consider using the image feature, the DNA feature, and their average as the key features for prediction. In all experiments, we use image features as the query. We

Table 1. Top-1 accuracy (%) on the validation set for different combinations of aligned embeddings (image, DNA, text) during contrastive training. We use **images** as queries and **DNA** as keys during inference. When aligning embeddings with text, we use a concatenation of the word representation of taxonomic levels. As a baseline, we show the results prior to contrastive learning (no alignment). We report the accuracy for seen and unseen species, and the harmonic mean (H.M.) between these (bold: highest acc.).

	Aligned embeddings			Micro top-1 acc			Macro top-1 acc		
Taxon	Img	DNA	Txt	Seen	Unseen	H.M.	Seen	Unseen	H.M.
Order				41.3	38.3	39.7	12.8	8.7	10.4
	✓	✓		98.5	**97.6**	98.0	87.4	52.6	68.7
	✓	✓	✓	**98.7**	**97.6**	**98.2**	**98.3**	**58.8**	**73.2**
Family				3.3	4.8	3.9	1.1	0.7	0.9
	✓	✓		80.4	74.0	77.1	50.4	28.3	36.2
	✓	✓	✓	**84.6**	**79.0**	**81.7**	**56.3**	**35.2**	**43.3**
Genus				0.9	3.1	1.4	0.1	0.1	0.1
	✓	✓		52.2	37.0	43.3	24.7	7.9	12.0
	✓	✓	✓	**58.5**	**43.5**	**49.9**	**30.1**	**11.7**	**16.9**
Species				0.0	2.0	0.0	0.0	0.0	0.0
	✓	✓		37.5	17.1	23.5	15.6	2.5	4.3
	✓	✓	✓	**42.0**	**30.1**	**35.1**	**17.4**	**3.9**	**6.4**

report the top-1 accuracy on the validation set in Table 1. All models are trained with a A100 GPU for 15 epochs with batch size of 400.

Taxonomic classification. As expected, the accuracy drops as we go from coarse (order) to fine-grained classification (species), and drop from seen to unseen species.

Are multimodal aligned embeddings useful? Our experiments show that using images, DNA barcodes, and text (taxonomic labels) together gives the better results in image classification compared with no finetuning and aligning with just text or DNA. This indicates that incorporating DNA barcodes and taxonomic labels into the learning process significantly enhances the model's ability to capture the representation of data.

What should we use as key? We can also see that when using DNA feature and averaged image and DNA feature as key, the accuracy dropped significantly. This suggests that the model does not perfectly align data across modalities.

4. Conclusion

We introduce BioScan-CLIP, an approach for integrating biological images with DNA barcodes and taxonomic labels to enhance species classification. Our experiments show that BioScan-CLIP learns and categorizes biological features effectively. Future efforts will aim at refining these techniques for broader application in biological research.

References

[1] C. Garcin, A. Joly, P. Bonnet, A. Affouard, J.-C. Lombardo, M. Chouet, M. Servajean, T. Lorieul, and J. Salmon. "Pl@ntNet-300K: a plant image dataset with high label ambiguity and a long-tailed distribution". In: *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*. Ed. by J. Vanschoren and S. Yeung. Vol. 1. Curran, 2021. URL: https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/7e7757b1e12abcb736ab9a754ffb617a-Paper-round2.pdf.

[2] G. Van Horn, O. Mac Aodha, Y. Song, Y. Cui, C. Sun, A. Shepard, H. Adam, P. Perona, and S. Belongie. "The iNaturalist Species Classification and Detection Dataset". In: *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Los Alamitos, CA, USA: IEEE Computer Society, 2018, pp. 8769–8778. DOI: 10.1109/CVPR.2018.00914.

[3] X.-S. Wei, Y.-Z. Song, O. M. Aodha, J. Wu, Y. Peng, J. Tang, J. Yang, and S. Belongie. "Fine-Grained Image Analysis With Deep Learning: A Survey". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44.12 (2022), pp. 8927–8948. DOI: 10.1109/TPAMI.2021.3126648.

[4] C. Martineau, D. Conte, R. Raveaux, I. Arnault, D. Munier, and G. Venturini. "A survey on image-based insect classification". In: *Pattern Recognition* 65 (2017), pp. 273–284. ISSN: 0031-3203. DOI: https://doi.org/10.1016/j.patcog.2016.12.020. URL: https://www.sciencedirect.com/science/article/pii/S0031320316304411.

[5] K. M. Ruppert, R. J. Kline, and M. S. Rahman. "Past, present, and future perspectives of environmental DNA (eDNA) metabarcoding: A systematic review in methods, monitoring, and applications of global eDNA". In: *Global Ecology and Conservation* 17 (2019), e00547. ISSN: 2351-9894. DOI: https://doi.org/10.1016/j.gecco.2019.e00547. URL: https://www.sciencedirect.com/science/article/pii/S2351989418303500.

[6] P. D. N. Hebert, A. Cywinska, S. L. Ball, and J. R. deWaard. "Biological identifications through DNA barcodes". In: *Proceedings of the Royal Society of London. Series B: Biological Sciences* 270.1512 (2003), pp. 313–321. DOI: 10.1098/rspb.2002.2218. eprint: https://royalsocietypublishing.org/doi/pdf/10.1098/rspb.2002.2218.

[7] Y. Ji, Z. Zhou, H. Liu, and R. V. Davuluri. "DNABERT: pre-trained Bidirectional Encoder Representations from Transformers model for DNA-language in genome". In: *Bioinformatics* 37.15 (2021), pp. 2112–2120.

[8] Z. Zhou, Y. Ji, W. Li, P. Dutta, R. Davuluri, and H. Liu. "DNABERT-2: Efficient foundation model and benchmark for multi-species genome". In: *arXiv preprint arXiv:2306.15006* (2023).

[9] F. Mock, F. Kretschmer, A. Kriese, S. Böcker, and M. Marz. "BERTax: taxonomic classification of DNA sequences with Deep Neural Networks". In: *BioRxiv* (2021), pp. 2021–07.

[10] S. Cahyawijaya, T. Yu, Z. Liu, T. T. Mak, X. Zhou, N. Y. Ip, and P. Fung. "SNP2Vec: Scalable Self-Supervised Pre-Training for Genome-Wide Association Study". In: *arXiv preprint arXiv:2204.06699* (2022).

[11] P. M. Arias, N. Sadjadi, M. Safari, Z. Gong, A. T. Wang, S. C. Lowe, J. B. Haurum, I. Zarubiieva, D. Steinke, L. Kari, et al. "BarcodeBERT: Transformers for Biodiversity Analysis". In: *arXiv preprint arXiv:2311.02401* (2023).

[12] S. Badirli, Z. Akata, G. Mohler, C. Picard, and M. M. Dundar. "Fine-grained zero-shot learning with dna as side information". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 19352–19362.

[13] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. "Learning transferable visual models from natural language supervision". In: *International conference on machine learning*. PMLR. 2021, pp. 8748–8763.

[14] S. Stevens, J. Wu, M. J. Thompson, E. G. Campolongo, C. H. Song, D. E. Carlyn, L. Dong, W. M. Dahdul, C. Stewart, T. Berger-Wolf, et al. "BioCLIP: A vision foundation model for the tree of life". In: *arXiv preprint arXiv:2311.18803* (2023).

[15] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. "LoRA: Low-rank adaptation of large language models". In: *arXiv preprint arXiv:2106.09685* (2021).

[16] K. Sohn. "Improved Deep Metric Learning with Multi-class N-pair Loss Objective". In: *Advances in Neural Information Processing Systems*. Ed. by D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett. Vol. 29. Curran Associates, Inc., 2016. URL: https://proceedings.

neurips.cc/paper_files/paper/2016/file/6b180037abbebea991d8b1232f8a8ca9-Paper.pdf.

[17] Y. Ruan, H.-H. Lee, Y. Zhang, K. Zhang, and A. X. Chang. "TriCoLo: Trimodal Contrastive Loss for Text To Shape Retrieval". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2024, pp. 5815–5825.

[18] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. "An image is worth 16x16 words: Transformers for image recognition at scale". In: *arXiv preprint arXiv:2010.11929* (2020).

[19] I. Turc, M.-W. Chang, K. Lee, and K. Toutanova. "Well-read students learn better: On the importance of pre-training compact models". In: *arXiv preprint arXiv:1908.08962* (2019).

[20] Z. Gharaee, Z. Gong, N. Pellegrino, I. Zarubiieva, J. B. Haurum, S. Lowe, J. McKeown, C. Ho, J. McLeod, Y.-Y. Wei, et al. "A Step Towards Worldwide Biodiversity Assessment: The BIOSCAN-1M Insect Dataset". In: *Advances in Neural Information Processing Systems* 36 (2024).